Jacobian-Based Stability Analysis of Human Posture

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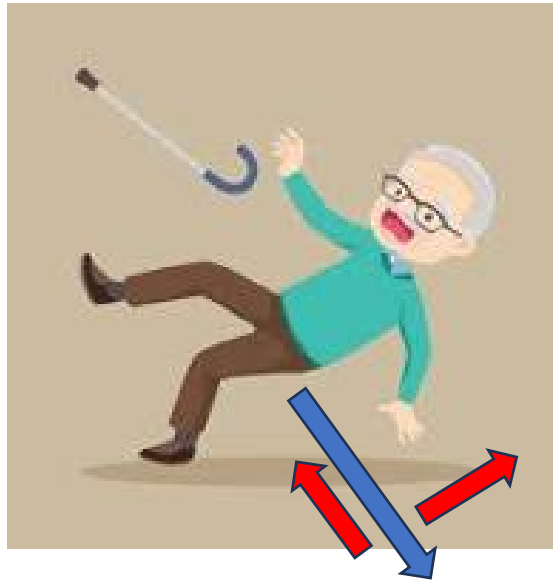
Problem Statement : Decoupling of Velocity Parameter

Decoupled Nature



Decoupled nature states that if body deviates in a particular direction, then corrective measure by the feedback system is in opposite direction.

Coupled Nature

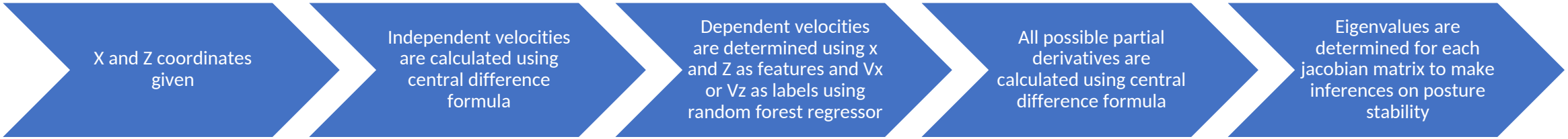


In coupled nature, even if the body deviates in a particular direction let say x axis, still feedback system makes corrective measures in both x and z axes.

Benefit of Coupling Approach

- Replicates the real scenario, where feedback corrections occur in a 2-D plane, in contrast to 1-D corrections in a decoupled state.
- In 1-D approach the values obtained for kinematic parameters will be overestimated, thus resulting in wrong postural analysis by the model. However, coupled state overcomes this shortcoming.
- Often in 1-D approach the summary metrics can be similar for 2 or more people, thus limiting only to broader classification of posture. In contrast, it is very rare to have similar values in coupled state due to internal variations in homeostatic mechanism.

Workflow Diagram



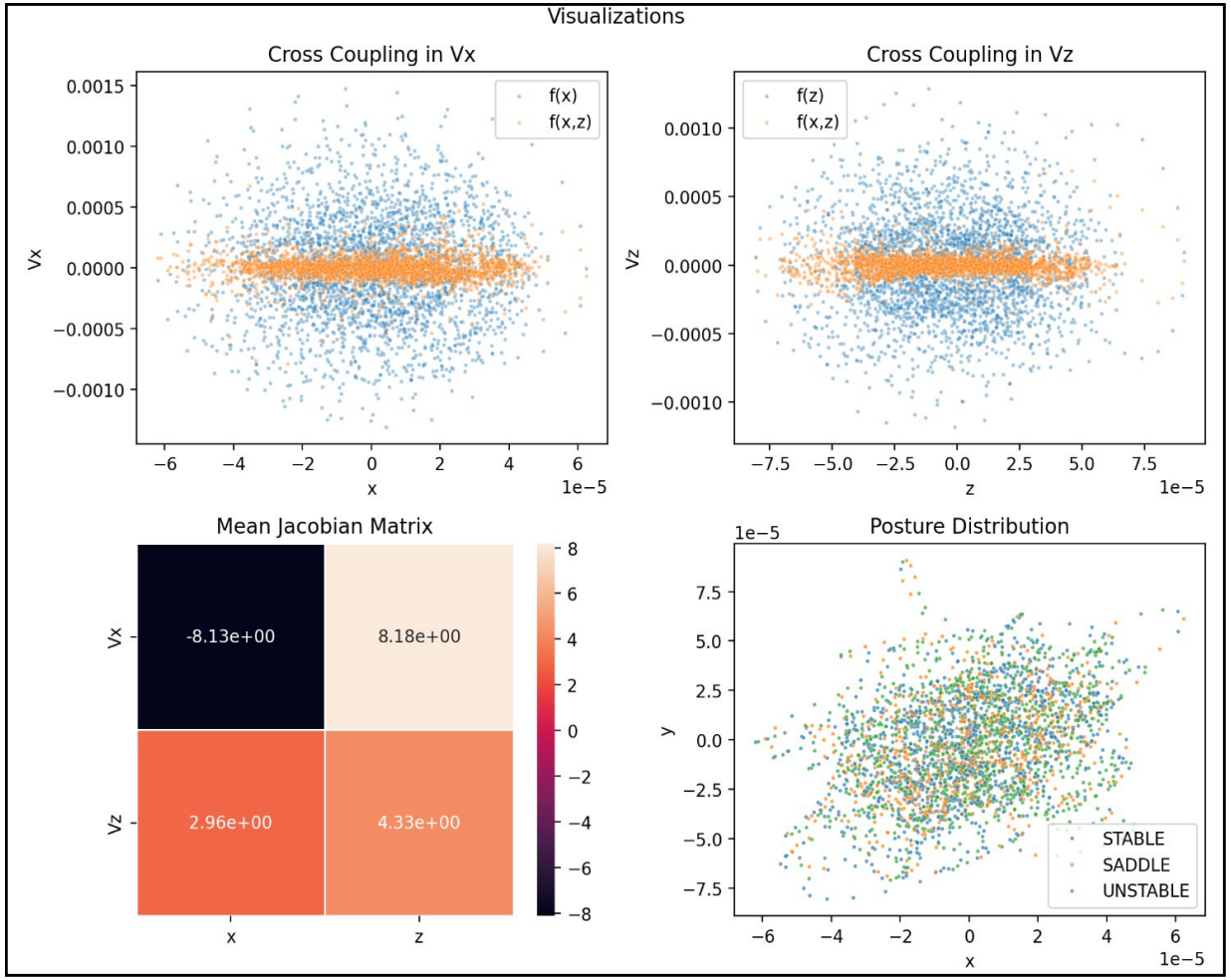
- The given process is performed for 2 patients (P1, P2).
- For each patient it is repeated 4 times to run the battery of tests in posturalography study (Eyes Open & Firm Surface, Eyes Closed & Firm Surface, Eyes Open and Foam Surface, Eyes Closed and Foam Surface).
- For each scenario visualizations have been created to prove the coupling of velocities and further the level of coupling.

Technical Details

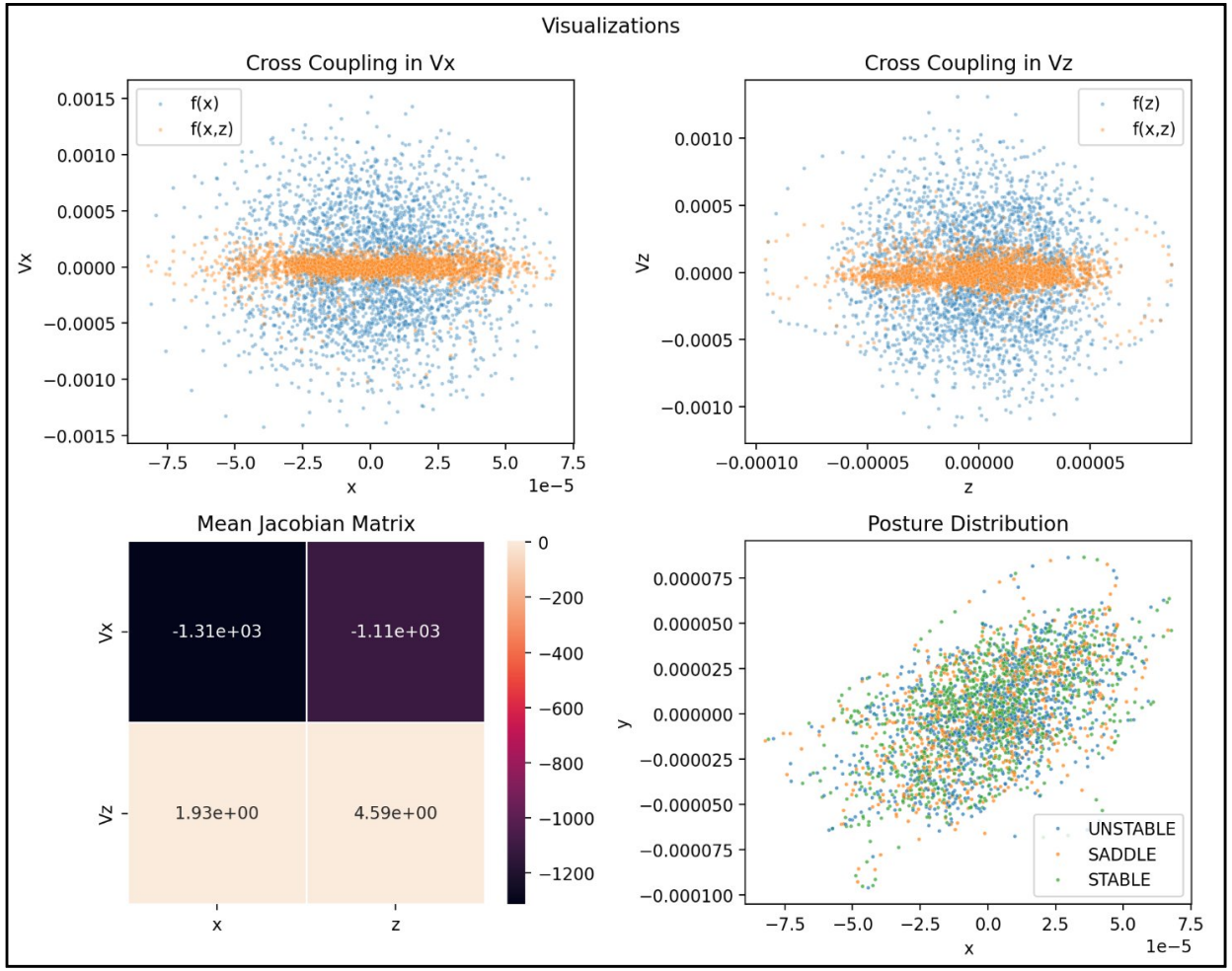
- X: coordinate in x axis, Z: coordinate in z axis
- Vx: independent velocity in x i.e. Vx(x), Vz: independent velocity in z i.e. Vz(z)
- dVx: dependent velocity in x i.e. Vx(x,z), dVz: dependent velocity in z i.e. Vz(x,z)
- Vx/X: partial derivative of dependent velocity in x with respect to x
- Vx/Z: partial derivative of dependent velocity in x with respect to z
- Vz/X: partial derivative of dependent velocity in z with respect to x
- Vz/Z: partial derivative of dependent velocity in z with respect to z
- Eigenvalues are calculated for each coordinate using the 2x2 jacobian matrix formed:
  - If both the eigenvalues have different sign it implies saddle/mixed posture
  - If both the eigenvalues have negative sign it implies stable posture
  - If both the eigenvalues have positive sign it implies unstable posture
  - If any eigenvalue is 0 then trace is used for posture type determination

Result Analysis

P1: Eyes Open & Firm Surface

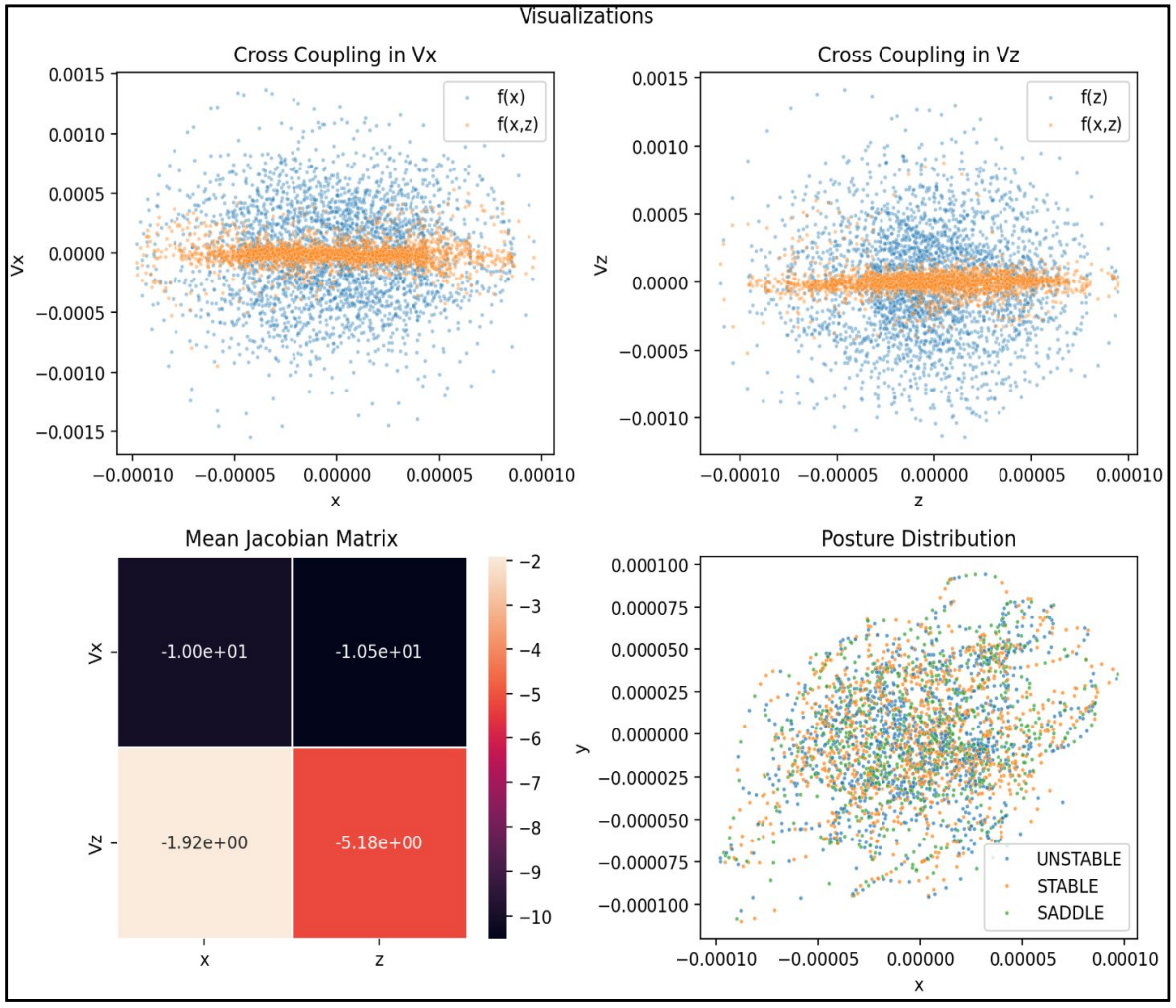


P1: Eyes Closed & Firm Surface

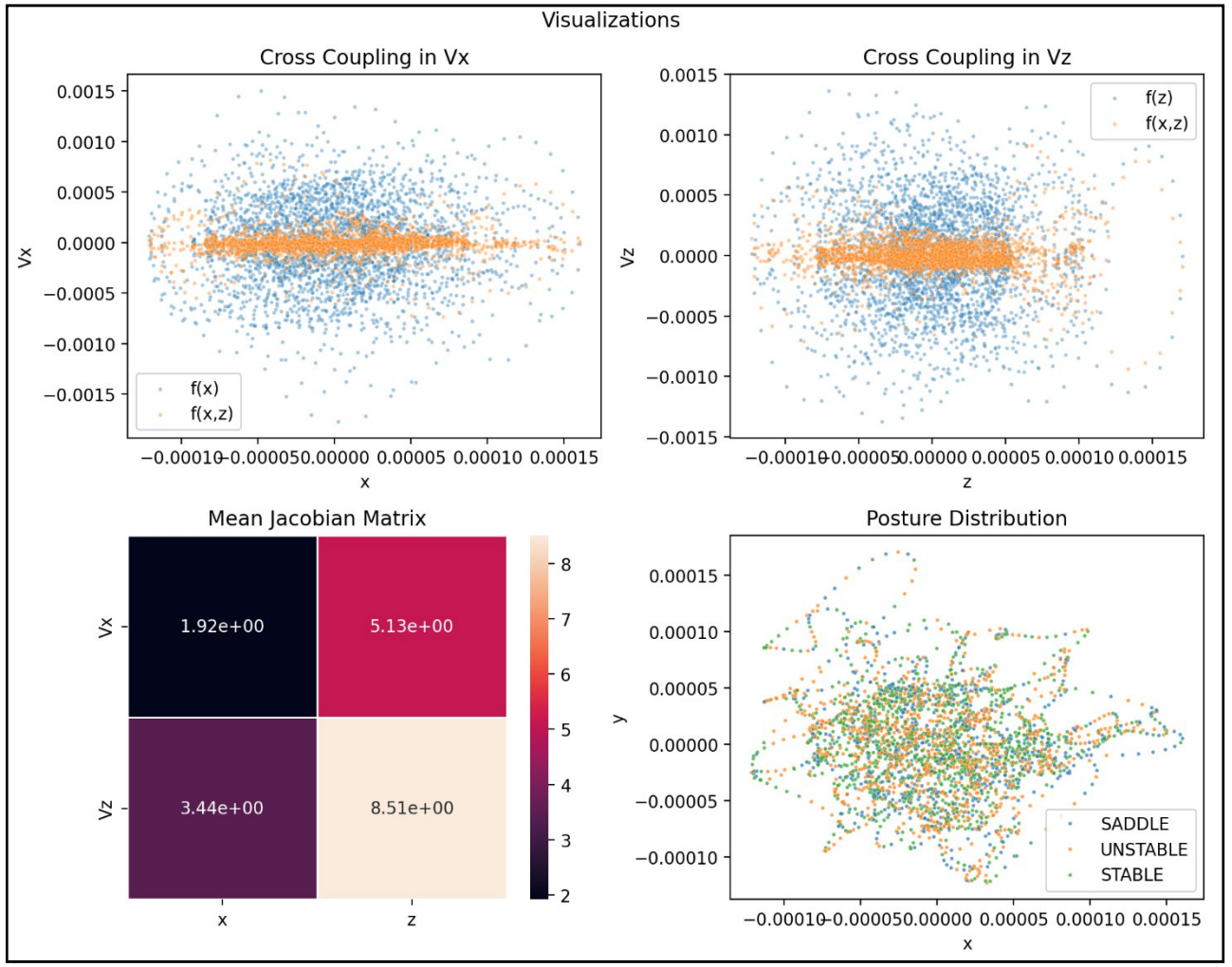


Result Analysis

P1: Eyes Open & Foam Surface

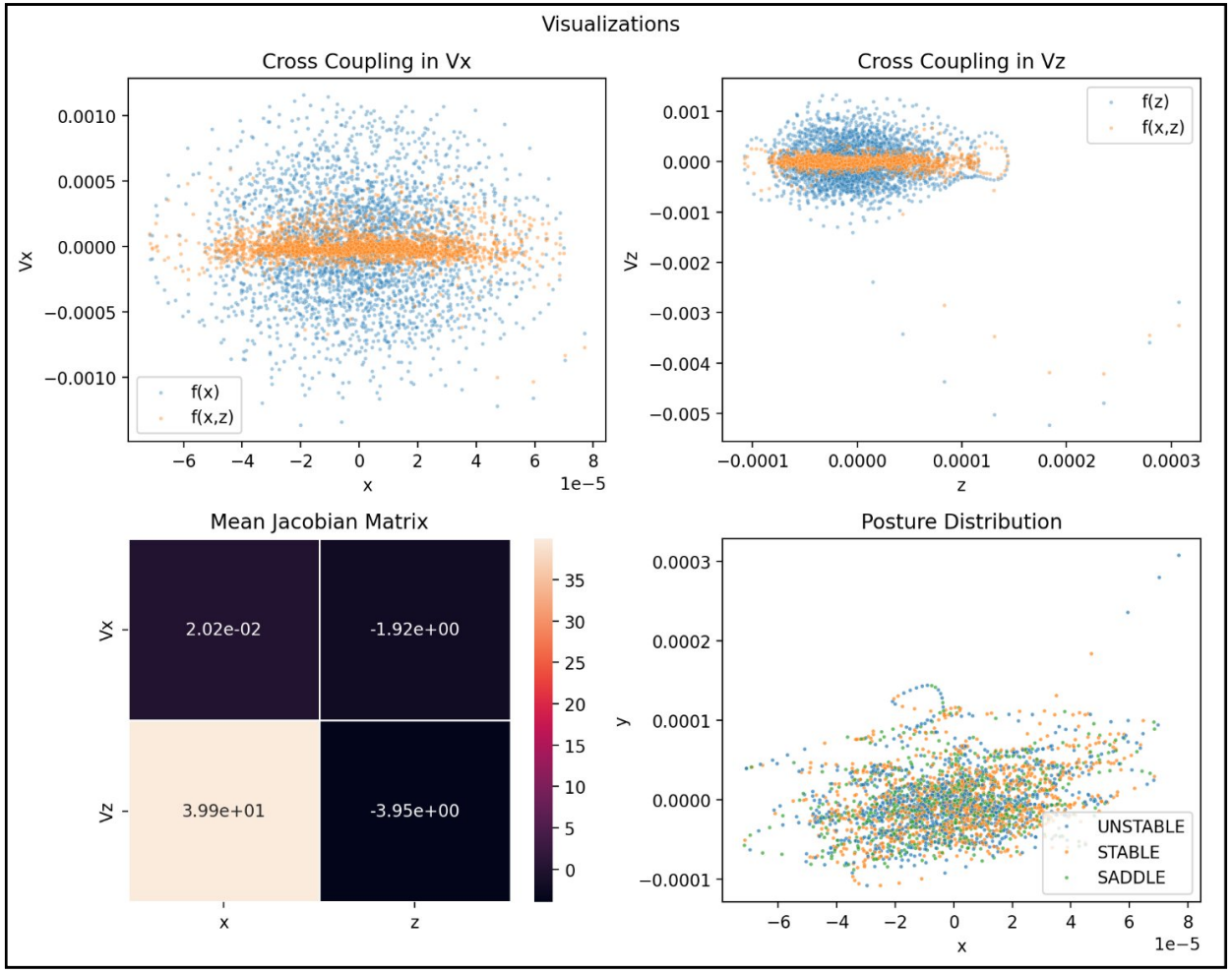


P1: Eyes Closed & Foam Surface

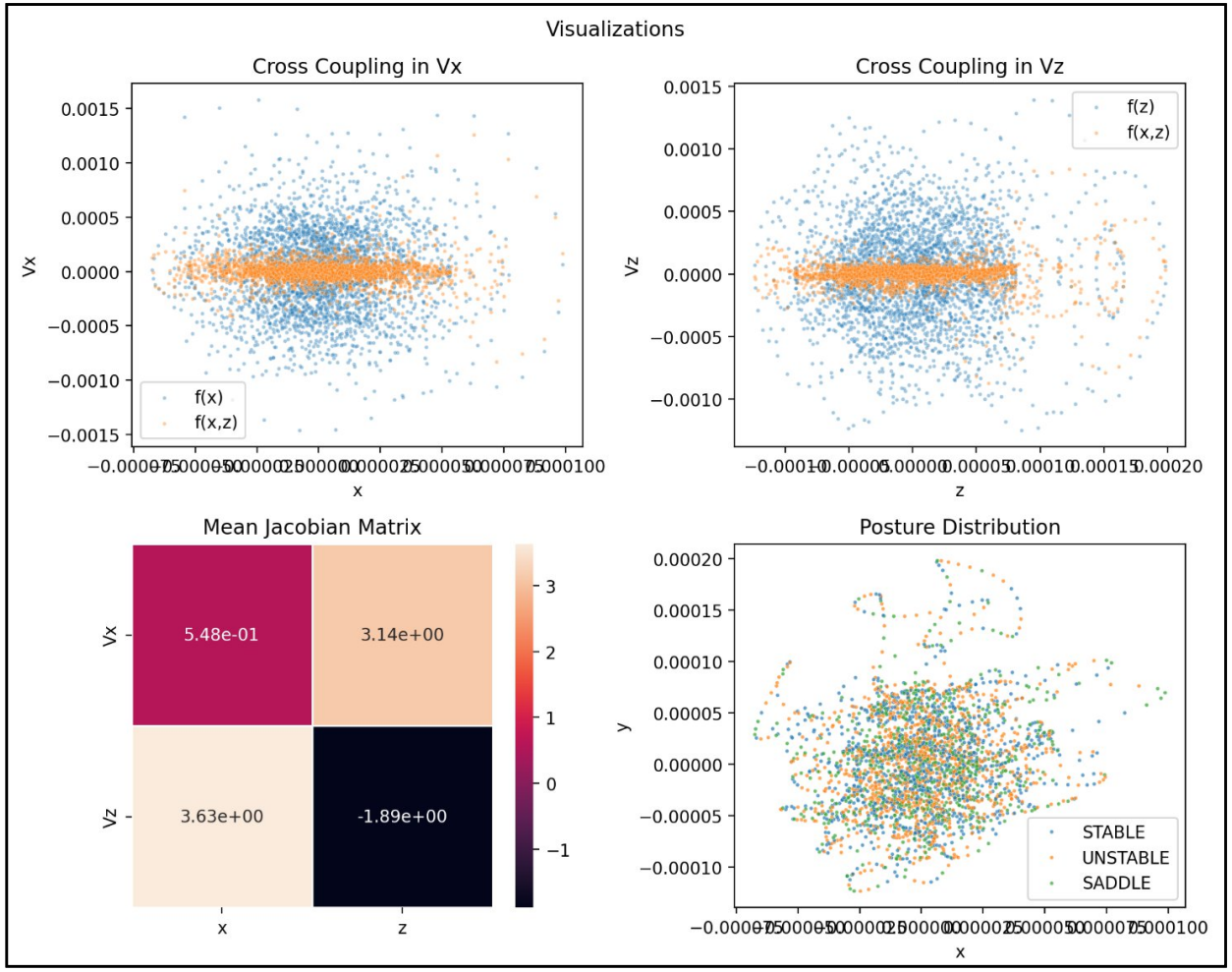


Result Analysis

P2: Eyes Open & Firm Surface

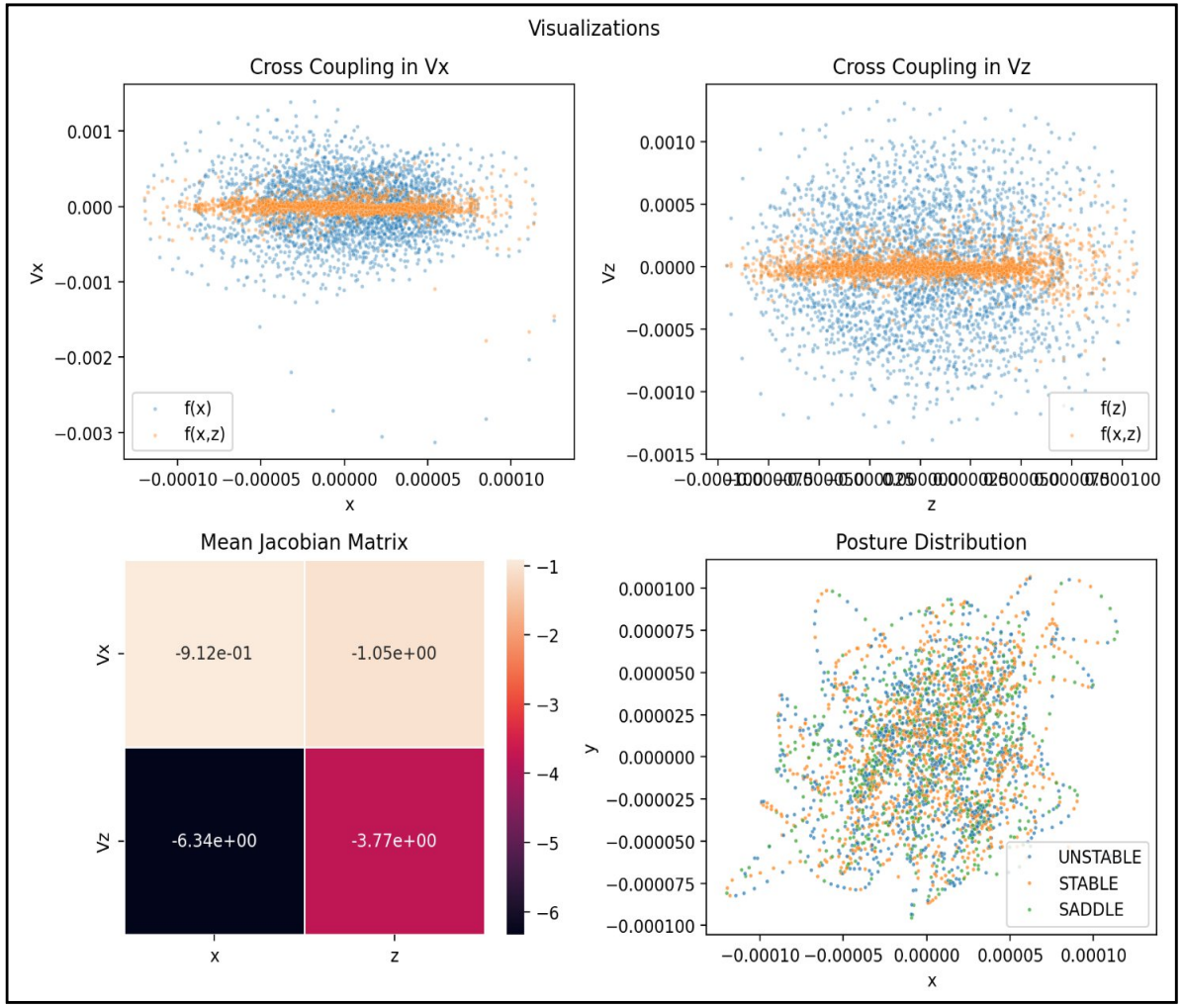


P2: Eyes Closed & Firm Surface

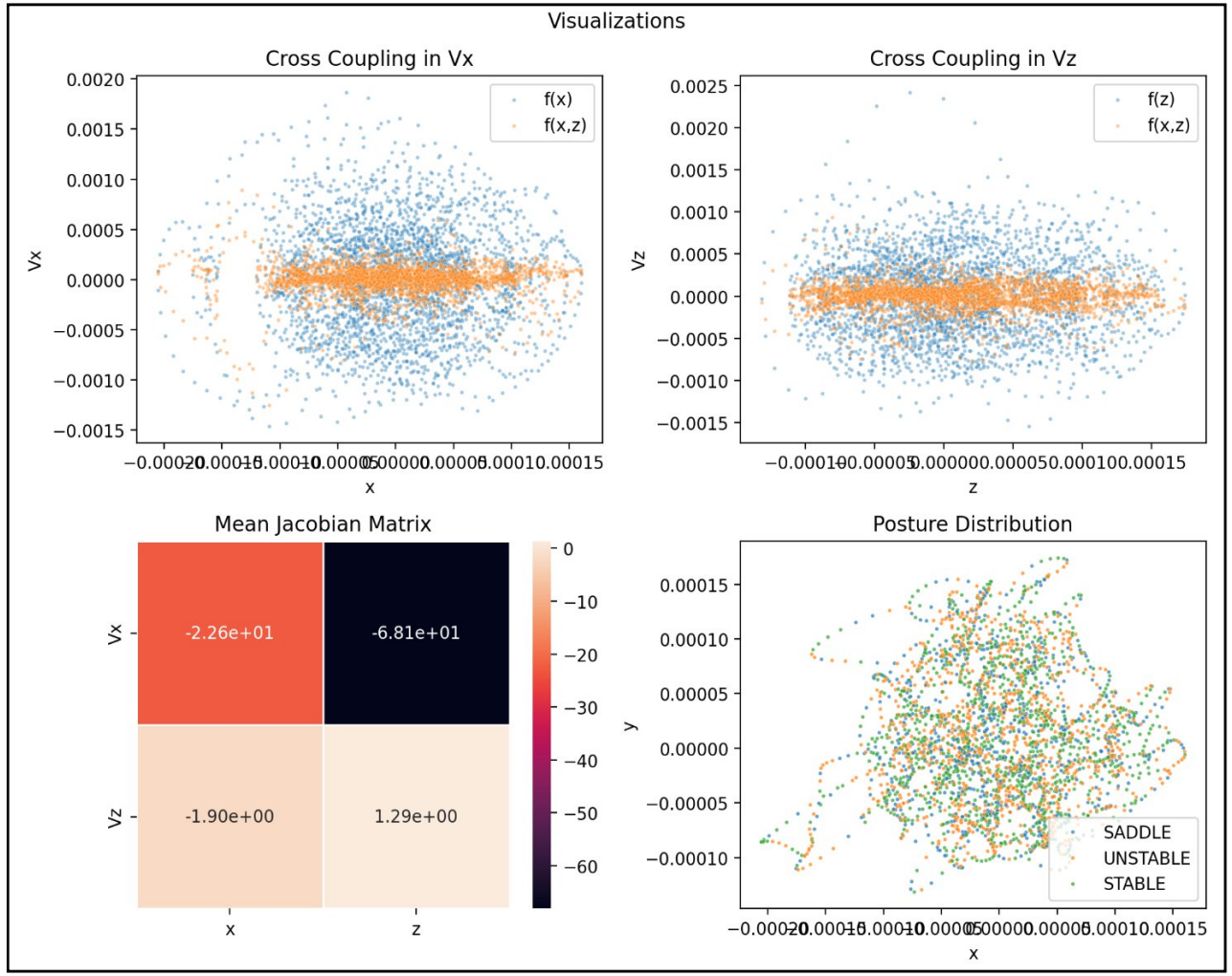


Result Analysis

P2: Eyes Open & Foam Surface



P2: Eyes Closed & Foam Surface



Clinical Test Thresholds Utilized

Test Type	System Tested
Eyes Open & Firm Surface	Standard
Eyes Closed & Firm Surface	Proprioception
Eyes Open & Foam Surface	Vision
Eyes Closed & Foam Surface	Vestibular Apparatus

Test	Normal	Physiological	Pathological
SOT, Modified CTSIB (Stable - Unstable)	>=30	10 – 29	<10
LOS, COP (Saddle)	>=720	650 - 719	<650

Comparative Analysis of 2 Patient

Patient	Test Type	Stable	Unstable	Stable - Unstable	Saddle	Inference
P1	Open+Firm	1127	1086	41	763	Normal + Normal
	Closed+Firm	1173	1058	115	745	Normal + Normal
	Open+Foam	1131	1107	24	738	Physiological + Normal
	Closed+Foam	1147	1119	28	710	Physiological + Physiological
P2	Open+Firm	1155	1104	51	717	Normal + Physiological
	Closed+Firm	1124	1113	11	739	Physiological + Normal
	Open+Foam	1162	1071	91	743	Normal + Normal
	Closed+Foam	1141	1105	36	730	Normal + Normal

Conclusion

- Following the above exploration, it can be confidently said that feedback mechanisms related to posture work in a 2-D plane in a coupled manner.
- Taking this approach helps medical professionals better analyze the posture status and make necessary corrective measures.
- If the final output is normal, nothing needs to be done. If the final output is physiological, then bodily changes like fatigue, anxiety, stress, and old age need to be ruled out. Finally, if the output is pathological, further system-specific tests are prescribed for further diagnosis.